Flap Angle Position Indicator – General data

1. General

A potentiometer is mounted on the port side fuselage attached to the inboard fore flap hinge arm at fuselage station 243.00, and sends a signal of the flap angle position through the DAU and onto the Flight Data Recorder and the flap angle position is indicated on the system screen.

2. Removal/Installation

Equipment and Materials:

- Corrosion preventative compound MIL–C–16173 Grade 2 or equivalent

A. Remove the Flap Angle Position Potentiometer

1) Ensure the electrical system is OFF (refer to 24-00-00, Electrical Power – General Data).

2) Pull the FLIGHT CONTROLS INDICATION (A14) circuit breaker and fit safety clip.

3) Disconnect the electrical connection.

4) Loosen the nut and screw that hold the lever assembly to the potentiometer shaft mounted in mounting bracket. Remove lever assembly.

5) Remove the safety wire and the (3) screws and clamps which secure the potentiometer onto the mounting bracket.

6) Remove potentiometer.

B. Install the Flap Angle Position Potentiometer

Note Coat all bolt shanks with the corrosive preventative compound MIL–C–16173 Grade 2 or equivalent before installation.

1) Ensure the electrical system is OFF (refer to 24-00-00, Electrical Power – General Data).

2) Ensure the FLIGHT CONTROLS INDICATION (A14) circuit breaker is pulled and fitted with safety clip.

3) Position the potentiometer on to the mounting bracket. Secure with the (3) clamps and screws. Safety wire screws.

4) Locate the lever assembly on potentiometer shaft so that the mechanism rotates freely. Ensure clearance is maintained with the structure through full travel, and a minimum gap of 0.05 inch between the lever assembly and the fuselage skin.

5) Secure with a washer, nut and screw.

6) Install bolt through the push rod assembly end, install between the bearing and the hinge arm, and bolt through the lever assembly end. Install washer, nut and cotter pin.

7) Adjust the flap angle position potentiometer. Refer to Para 3.A.

8) Reconnect the electrical connection.

9) Replace FLIGHT CONTROLS INDICATION (A14) circuit breaker.

3. Adjustment/Test

A. Adjustment Flap Angle Position Potentiometer

Equipment and Materials:

- DC Voltmeter
- Digital Inclinometer

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Measure the Ground Power Voltage (V_{GP}) at the external power relay (or other suitable location).

3) Pull the FLIGHT CONTROLS INDICATION (A14) circuit breaker and fit with safety clip.

4) Select flaps down 20°.

5) Check that the flaps have reached 20° with the digital inclinometer. Move the flap selector up or down to get exactly 20° with inclinometer.

6) Disconnect terminal 2 of the electrical connection going to the potentiometer and connect voltmeter positive to the lead from terminal 2 while attaching voltmeter negative to aircraft ground.

CAUTION **ENSURE THE TERMINAL 2 LEAD DOES NOT COME INTO CONTACT WITH AN ELECTRICAL GROUND.**

7) Loosen screw and nut on the lever assembly at the potentiometer shaft enough that the shaft turns freely.

8) Reset the FLIGHT CONTROLS INDICATION (A14) circuit breaker.

9) Rotate potentiometer shaft until the measured DC Voltage is precisely 50% of total aircraft voltage V_{GP} (approximately 14 volts).

10) Pull FLIGHT CONTROLS INDICATION (A14) circuit breaker, fit safety clip and disconnect voltmeter from potentiometer. Tighten hardware on lever at potentiometer shaft.

Note Do not rotate potentiometer shaft during this process.

11) Reconnect the potentiometer electrical connection.

12) Remove safety clip and reset FLIGHT CONTROLS INDICATION (A14) circuit breaker.

13) Do a function test. (Refer to Para 3.B).

Note For Flight Data Recorder input checks, refer to 31-31-00, Flight Data Recorder (FDR) – General Data.

For Data Acquisition Unit input checks, refer to 31-44-00, Data Acquisition Unit (DAU) (Mod 6/2049) – General Data.

B. Function Test Flap Angle Position Potentiometer

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Wait several minutes for the aircraft electrical and avionics systems to initialize.

3) Deflect flaps to the full up position and note flap position display indication correlates to 0°.

4) Deflect flaps to the full down position (Reference inboard fore flap 40°) and note the flap position display indication correlates to 37°.

5) If the flap display indication does not match the flap position, repeat adjustment steps in Para 3.A.

6) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

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